🧭 M003 Guide — Counting (x, y, z)

Topic: Before 3D · **Course:** 3D Coordinates · **Level:** All · **Time:** about 20 minutes

Pixel art used **two** numbers. A 3D build needs **three**. They always come in the same order:

1st = X	2nd = Y	3rd = Z
across →	up ↑	forward ⤴
starts at 1	starts at 0	starts at 1

Count X, then Y, then Z. Always that order.

> 🔴 Stand on your **home spot** (red block) every run. Move your feet, move your build.

1 · Build Your Grid 🎨

Make a striped ruler for each axis, so you can **see** the numbers.

Stand on your home spot. Run this to lay the white floor and the yellow lines:

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
        pos(1, -1, 1),
        pos(15, -1, 15),
        FillOperation.REPLACE)
    blocks.fill(YELLOW_CONCRETE,
        pos(5, -1, 1),
        pos(5, -1, 15),
        FillOperation.REPLACE)
    blocks.fill(YELLOW_CONCRETE,
        pos(10, -1, 1),
        pos(10, -1, 15),
        FillOperation.REPLACE)
    blocks.fill(YELLOW_CONCRETE,
        pos(1, -1, 5),
        pos(15, -1, 5),
        FillOperation.REPLACE)
    blocks.fill(YELLOW_CONCRETE,
        pos(1, -1, 10),
        pos(15, -1, 10),
        FillOperation.REPLACE)
player.on_chat("grid", on_run)
```

Now place the three rulers by hand — **yellow, black, yellow, black** all the way:

- **X ruler** — walk **right**, one block at a time.
- **Z ruler** — walk **forward**, one block at a time.
- **Y ruler** — stack a pillar **up** from the corner.

> 🟨⬛ The stripes are there so you can count without guessing.

2 · Count X First ➡️

X is across. Follow the striped line going left to right.

The block **next to** your home spot is **1**. Then 2, 3, 4 ...

```
place red block at (1, ?, ?)
place red block at (2, ?, ?)
place red block at (3, ?, ?)
```

Which way do these blocks go? Circle one: across · up · forward

X starts at which number? Circle one: 0 · 1

3 · Then Count Y ⬆️

Y is up. Follow the striped pillar.

⚠️ **Y is different.** The block on the **ground** is **0** — not 1. Then 1, 2, 3 ...

```
place red block at (?, 0, ?)   # on the ground
place red block at (?, 1, ?)   # one higher
place red block at (?, 2, ?)   # one higher again
```

Y starts at which number? Circle one: 0 · 1

A block at y = 3 is ...? Circle one: on the ground · 3 blocks up

4 · Last, Count Z ⤴️

Z is forward — away from you, deeper into the grid.

The block **in front of** your home spot is **1**. Then 2, 3, 4 ...

```
place red block at (?, ?, 1)
place red block at (?, ?, 2)
place red block at (?, ?, 3)
```

Which number is like x — starts at 1? Circle one: y · z

5 · Put Them Together 🧭

Read **(5, 2, 3)** in three steps:

Step	Number	Ruler	Do this
1	x = 5	across	walk 5 right
2	y = 2	up	go 2 up (ground is 0)
3	z = 3	forward	go 3 forward

Where is (4, 0, 6)? Circle one: on the ground · 4 blocks up · 6 blocks up

Which coordinate is on the ground? Circle one: (2, 1, 2) · (2, 0, 2)

These two blocks are stacked. Which number grew?

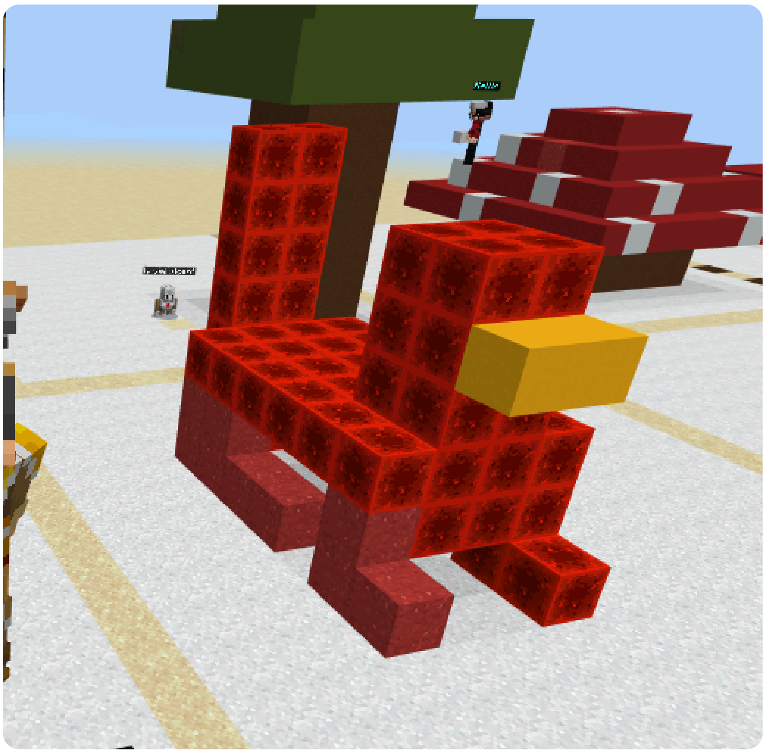
```
place red block at (7, 1, 2)
place red block at (7, 2, 2)
```

Circle one: x · y · z

6 · Show Your Work 📸 🎥

Warm-up: stand on your home spot and place one block at **(3, 0, 3)**. Check it against your rulers.

Build the chicken. Count x, then y, then z for every block.



> 🧱 Recipe (red nether brick, yellow concrete for the beak):
>
> - legs: (4, 0, 4) and (4, 0, 5)
> - body: (3, 1, 4) to (6, 2, 5)
> - tail: (2, 3, 4) and (2, 3, 5)
> - head: (6, 3, 4) to (7, 4, 5)
> - beak: yellow at (8, 3, 4) and (8, 3, 5)

If you finish early, give it a comb on top of the head.

Record **one video** (a phone is fine). Show two things:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Count one block out loud — x, then y, then z.

Say these out loud, filling in the blanks:

> Today I built _____.
>
> I built it using this code: _____.

> In this code I used ______.
>
> The hardest part was ______.
>
> That part was hard because ______.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.